
```
1 // Prints an integer
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     int n = 50;
8     printf("%i\n", n);
9 }
```

```
1  // Prints an integer's address
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int n = 50;
8      printf("%p\n", &n);
9  }
```

```
1  // Stores and prints an integer's address
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int n = 50;
8      int *p = &n;
9      printf("%p\n", p);
10 }
```

```
1  // Stores and prints an integer via its address
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      int n = 50;
8      int *p = &n;
9      printf("%i\n", *p);
10 }
```

```
1  // Prints a string
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      string s = "HI!";
9      printf("%s\n", s);
10 }
```

```
1 // Prints a string's address as well the addresses of its chars
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     string s = "HI!";
9     printf("%p\n", s);
10    printf("%p\n", &s[0]);
11    printf("%p\n", &s[1]);
12    printf("%p\n", &s[2]);
13    printf("%p\n", &s[3]);
14 }
```

```
1  // Declares a string with CS50 Library
2
3  #include <cs50.h>
4  #include <stdio.h>
5
6  int main(void)
7  {
8      string s = "HI!";
9      printf("%s\n", s);
10 }
```

```
1 // Declares a string without CS50 Library
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char *s = "HI!";
8     printf("%s\n", s);
9 }
```

```
1  // Prints a string's chars
2
3  #include <stdio.h>
4
5  int main(void)
6  {
7      char *s = "HI!";
8      printf("%c\n", s[0]);
9      printf("%c\n", s[1]);
10     printf("%c\n", s[2]);
11 }
```

```
1 // Prints a string's chars via pointer arithmetic
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char *s = "HI!";
8     printf("%c\n", *s);
9     printf("%c\n", *(s + 1));
10    printf("%c\n", *(s + 2));
11 }
```

```
1 // Prints substrings via pointer arithmetic
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char *s = "HI!";
8     printf("%s\n", s);
9     printf("%s\n", s + 1);
10    printf("%s\n", s + 2);
11 }
```

```
1 // Compares two integers
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Get two integers
9     int i = get_int("i: ");
10    int j = get_int("j: ");
11
12    // Compare integers
13    if (i == j)
14    {
15        printf("Same\n");
16    }
17    else
18    {
19        printf("Different\n");
20    }
21 }
```

```
1 // Compares two strings' addresses
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Get two strings
9     char *s = get_string("s: ");
10    char *t = get_string("t: ");
11
12    // Compare strings' addresses
13    if (s == t)
14    {
15        printf("Same\n");
16    }
17    else
18    {
19        printf("Different\n");
20    }
21 }
```

```
1  // Compares two strings using strcmp
2
3  #include <cs50.h>
4  #include <stdio.h>
5  #include <string.h>
6
7  int main(void)
8  {
9      // Get two strings
10     char *s = get_string("s: ");
11     char *t = get_string("t: ");
12
13     // Compare strings
14     if (strcmp(s, t) == 0)
15     {
16         printf("Same\n");
17     }
18     else
19     {
20         printf("Different\n");
21     }
22 }
```

```
1 // Prints two strings
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Get two strings
9     char *s = get_string("s: ");
10    char *t = get_string("t: ");
11
12    // Print strings
13    printf("%s\n", s);
14    printf("%s\n", t);
15 }
```

```
1 // Prints two strings' addresses
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     // Get two strings
9     char *s = get_string("s: ");
10    char *t = get_string("t: ");
11
12    // Print strings' addresses
13    printf("%p\n", s);
14    printf("%p\n", t);
15 }
```

```
1 // Capitalizes a string
2
3 #include <cs50.h>
4 #include <ctype.h>
5 #include <stdio.h>
6 #include <string.h>
7
8 int main(void)
9 {
10     // Get a string
11     string s = get_string("s: ");
12
13     // Copy string's address
14     string t = s;
15
16     // Capitalize first letter in string
17     t[0] = toupper(t[0]);
18
19     // Print string twice
20     printf("s: %s\n", s);
21     printf("t: %s\n", t);
22 }
```

```
1 // Capitalizes a string, checking length first
2
3 #include <cs50.h>
4 #include <ctype.h>
5 #include <stdio.h>
6 #include <string.h>
7
8 int main(void)
9 {
10     // Get a string
11     string s = get_string("s: ");
12
13     // Copy string's address
14     string t = s;
15
16     // Capitalize first letter in string
17     if (strlen(t) > 0)
18     {
19         t[0] = toupper(t[0]);
20     }
21
22     // Print string twice
23     printf("s: %s\n", s);
24     printf("t: %s\n", t);
25 }
```

```
1  // Capitalizes a copy of a string
2
3  #include <cs50.h>
4  #include <ctype.h>
5  #include <stdio.h>
6  #include <stdlib.h>
7  #include <string.h>
8
9  int main(void)
10 {
11     // Get a string
12     char *s = get_string("s: ");
13
14     // Allocate memory for another string
15     char *t = malloc(strlen(s) + 1);
16
17     // Copy string into memory, including '\0'
18     for (int i = 0; i <= strlen(s); i++)
19     {
20         t[i] = s[i];
21     }
22
23     // Capitalize copy
24     t[0] = toupper(t[0]);
25
26     // Print strings
27     printf("s: %s\n", s);
28     printf("t: %s\n", t);
29 }
```

```
1 // Capitalizes a copy of a string, defining n in loop too
2
3 #include <cs50.h>
4 #include <ctype.h>
5 #include <stdio.h>
6 #include <stdlib.h>
7 #include <string.h>
8
9 int main(void)
10 {
11     // Get a string
12     char *s = get_string("s: ");
13
14     // Allocate memory for another string
15     char *t = malloc(strlen(s) + 1);
16
17     // Copy string into memory, including '\0'
18     for (int i = 0, n = strlen(s); i <= n; i++)
19     {
20         t[i] = s[i];
21     }
22
23     // Capitalize copy
24     t[0] = toupper(t[0]);
25
26     // Print strings
27     printf("s: %s\n", s);
28     printf("t: %s\n", t);
29 }
```

```
1  // Capitalizes a copy of a string using strcpy
2
3  #include <cs50.h>
4  #include <ctype.h>
5  #include <stdio.h>
6  #include <stdlib.h>
7  #include <string.h>
8
9  int main(void)
10 {
11     // Get a string
12     char *s = get_string("s: ");
13
14     // Allocate memory for another string
15     char *t = malloc(strlen(s) + 1);
16
17     // Copy string into memory
18     strcpy(t, s);
19
20     // Capitalize copy
21     t[0] = toupper(t[0]);
22
23     // Print strings
24     printf("s: %s\n", s);
25     printf("t: %s\n", t);
26 }
```

```
1 // Capitalizes a copy of a string without memory errors
2
3 #include <cs50.h>
4 #include <ctype.h>
5 #include <stdio.h>
6 #include <stdlib.h>
7 #include <string.h>
8
9 int main(void)
10 {
11     // Get a string
12     char *s = get_string("s: ");
13     if (s == NULL)
14     {
15         return 1;
16     }
17
18     // Allocate memory for another string
19     char *t = malloc(strlen(s) + 1);
20     if (t == NULL)
21     {
22         return 1;
23     }
24
25     // Copy string into memory
26     strcpy(t, s);
27
28     // Capitalize copy
29     if (strlen(t) > 0)
30     {
31         t[0] = toupper(t[0]);
32     }
33
34     // Print strings
35     printf("s: %s\n", s);
36     printf("t: %s\n", t);
37
38     // Free memory
39     free(t);
40     return 0;
41 }
```

```
1  // Demonstrates memory errors via valgrind
2
3  #include <stdio.h>
4  #include <stdlib.h>
5
6  int main(void)
7  {
8      int *x = malloc(3 * sizeof(int));
9      x[1] = 72;
10     x[2] = 73;
11     x[3] = 33;
12 }
```

```
1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int main(void)
5  {
6      int scores[1024];
7      for (int i = 0; i < 1024; i++)
8      {
9          printf("%i\n", scores[i]);
10     }
11 }
```

```
1 // Fails to swap two integers
2
3 #include <stdio.h>
4
5 void swap(int a, int b);
6
7 int main(void)
8 {
9     int x = 1;
10    int y = 2;
11
12    printf("x is %i, y is %i\n", x, y);
13    swap(x, y);
14    printf("x is %i, y is %i\n", x, y);
15 }
16
17 void swap(int a, int b)
18 {
19     int tmp = a;
20     a = b;
21     b = tmp;
22 }
```

```
1  // Swaps two integers using pointers
2
3  #include <stdio.h>
4
5  void swap(int *a, int *b);
6
7  int main(void)
8  {
9      int x = 1;
10     int y = 2;
11
12     printf("x is %i, y is %i\n", x, y);
13     swap(&x, &y);
14     printf("x is %i, y is %i\n", x, y);
15 }
16
17 void swap(int *a, int *b)
18 {
19     int tmp = *a;
20     *a = *b;
21     *b = tmp;
22 }
```

```
1 // Gets an int from user using get_int
2
3 #include <cs50.h>
4 #include <stdio.h>
5
6 int main(void)
7 {
8     int n = get_int("n: ");
9     printf("n: %i\n", n);
10 }
```

```
1 // Gets an int from user using scanf
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     int n;
8     printf("n: ");
9     scanf("%i", &n);
10    printf("n: %i\n", n);
11 }
```

```
1 // Incorrectly gets a string from user using scanf; compile with -Wno-uninitialized
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char *s;
8     printf("s: ");
9     scanf("%s", s);
10    printf("s: %s\n", s);
11 }
```

```
1 // Dangerously gets a string from user using scanf with array
2
3 #include <stdio.h>
4
5 int main(void)
6 {
7     char s[4];
8     printf("s: ");
9     scanf("%s", s);
10    printf("s: %s\n", s);
11 }
```

```
1 // Saves names and numbers to a CSV file
2
3 #include <cs50.h>
4 #include <stdio.h>
5 #include <string.h>
6
7 int main(void)
8 {
9     // Open CSV file
10    FILE *file = fopen("phonebook.csv", "a");
11
12    // Get name and number
13    char *name = get_string("Name: ");
14    char *number = get_string("Number: ");
15
16    // Print to file
17    fprintf(file, "%s,%s\n", name, number);
18
19    // Close file
20    fclose(file);
21 }
```

```
1 // Saves names and numbers to a CSV file, checking for NULL
2
3 #include <cs50.h>
4 #include <stdio.h>
5 #include <string.h>
6
7 int main(void)
8 {
9     // Open CSV file
10    FILE *file = fopen("phonebook.csv", "a");
11    if (file == NULL)
12    {
13        return 1;
14    }
15
16    // Get name and number
17    char *name = get_string("Name: ");
18    char *number = get_string("Number: ");
19
20    // Print to file
21    fprintf(file, "%s,%s\n", name, number);
22
23    // Close file
24    fclose(file);
25 }
```

```
1 // Copies a file
2
3 #include <stdio.h>
4
5 typedef unsigned char BYTE;
6
7 int main(int argc, char *argv[])
8 {
9     FILE *src = fopen(argv[1], "rb");
10    FILE *dst = fopen(argv[2], "wb");
11
12    BYTE b;
13
14    while (fread(&b, sizeof(b), 1, src) != 0)
15    {
16        fwrite(&b, sizeof(b), 1, dst);
17    }
18
19    fclose(dst);
20    fclose(src);
21 }
```